

Olist E-commerce Data Analysis

Objective:

Analyze Brazilian e-commerce performance across revenue, products, sellers, customers, delivery operations, cancellations, and customer satisfaction using PostgreSQL and Power BI.

Tools:

PostgreSQL (SQL) • Power BI • LibreOffice Writer

Analysis Structure:

Financial / Commercial Performance • Product Performance & Demand • Seller Performance • Customer Retention • Order & Delivery Operations • Cancellations / Unsuccessful Orders • Customer Satisfaction

Methodology

The analysis followed a structured process:

Business Question → Data Preparation & Quality Checks → KPI Definition → SQL Analysis in PostgreSQL → Power BI Visualization → Business Insights & Recommendations

Key metric definitions:

- **Revenue** = price + freight_value
- **Repeat customer** = customer_unique_id with more than 1 delivered order
- **One-time customer** = customer_unique_id with exactly 1 delivered order
- **Cancellation rate** = cancelled orders ÷ eligible orders
- **Low review score** = review score 1–3
- **High review score** = review score 4–5

1. Revenue & Commercial Performance

Finding: The Olist dataset generated approximately **R\$15.37M in total revenue**, with revenue concentrated among a relatively small group of product categories. The leading category, **health_beauty**, generated the highest revenue among the top 15 categories.

Business implication: Overall revenue is not evenly distributed across product categories, so performance changes in the leading categories can have a meaningful effect on total revenue.

Recommendation: Monitor the leading revenue-generating categories closely while maintaining a broader category mix to reduce dependence on a small number of categories.

2. Order & Delivery Performance

Finding: Delivery performance is an important operational factor in the customer experience. The dashboard shows that most orders were **On-Time / Early**, while a smaller share were **Late**. The delivery-performance comparison also shows a substantial difference in average review scores between the two groups.

Business implication: Late deliveries are associated with a weaker customer experience, making delivery reliability an important operational performance area.

Recommendation: Monitor late-delivery rates regularly and investigate the products, sellers, and locations with consistently higher late-delivery rates.

3. Customer Retention

Finding: Repeat customers represent only **3.0% of the customer base**, while one-time customers make up the vast majority. The purchase-interval analysis also shows that repeat purchases occur across several time intervals rather than within a single consistent window.

Business implication: Customer acquisition is much more prevalent than repeat purchasing, indicating that retaining existing customers is an important area for further analysis.

Recommendation: Investigate which products, categories, and customer segments are associated with repeat purchasing, and use the purchase-interval patterns to inform retention campaigns.

4. Cancellations & Unsuccessful Orders

Finding: The overall cancellation rate was approximately **0.62%**, indicating that cancellations represented a relatively small share of eligible orders. However, cancellation rates varied across months, product categories, sellers, and geographic areas. The monthly trend shows some periods exceeding **1%**.

Business implication: The overall rate can appear stable while specific products, sellers, or locations experience substantially higher cancellation levels.

Recommendation: Monitor cancellation rate using minimum order-volume thresholds and investigate consistently high-rate products, sellers, and geographic areas rather than focusing only on the overall average.

5. Seller & Product Contribution

Finding: Seller and product revenue contributions vary substantially. The dashboard shows a distinct group of top sellers and top products generating the highest revenue, while many other sellers and products contribute much smaller amounts.

Business implication: Overall commercial performance depends partly on a relatively small set of high-contributing sellers and products.

Recommendation: Monitor high-contribution sellers and products regularly while also evaluating lower-volume sellers and products using appropriate minimum-volume thresholds before making performance comparisons.

6. Customer Satisfaction

Finding: Average review scores were **4.15 for one-time customers** and **4.21 for repeat customers**. The low-score share was **21.15% for one-time customers** and **20.07% for repeat customers**.

Business implication: The repeat-customer group had a slightly higher average review score and a slightly lower share of low scores, although the difference was relatively small.

Recommendation: Monitor satisfaction across customer types and investigate the specific product and delivery factors associated with low review scores.

7. Data Quality & Analysis Limitations

Finding: Several data-quality issues were identified during the analysis, including missing delivery timestamps, anomalous delivery timestamps, and potential row multiplication when combining order-level data with payments or reviews.

Business implication: Some operational metrics, especially delivery-timing measures, can be affected by incomplete or inconsistent timestamps.

Recommendation: Apply explicit data-quality rules for each metric, document exclusions and assumptions, and avoid combining tables at incompatible grains without appropriate aggregation.

Overall Conclusion

The analysis shows a business with strong overall order and revenue performance, while several areas require closer attention. Revenue is concentrated across leading product categories, repeat customers represent a small share of the customer base, and delivery delays are associated with lower review scores. Although the overall cancellation rate is low, variation exists across products, sellers, months, and geographic areas. These findings provide clear areas for improving retention, delivery reliability, and targeted performance monitoring.